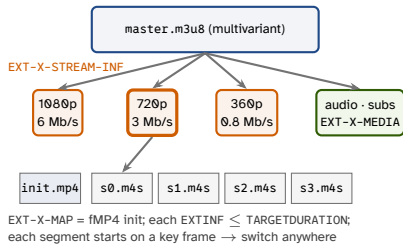
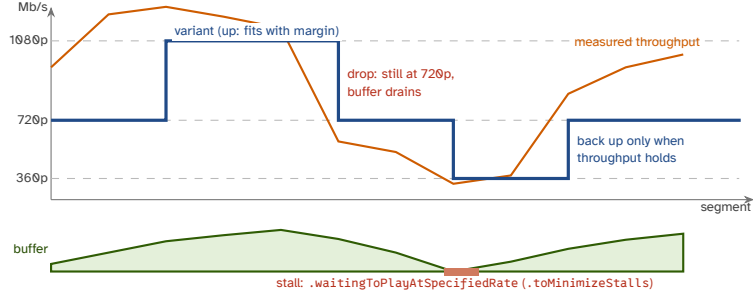


In one line: **HLS** = plain HTTP files: a **multivariant playlist** lists the same video at several bitrates, each variant is a **media playlist** of short **segments** (.ts or fMP4). The player measures throughput and buffer and **switches variant at segment boundaries** (ABR). In AVFoundation you watch **state** — `AVPlayerItem.status`, `AVPlayer.timeControlStatus` + `reasonForWaitingToPlay` — and tune **startup, caps and buffer**, never the download itself.

Playlist hierarchy



ABR over time — throughput, chosen variant, buffer



HLS (RFC 8216)

- Multivariant** playlist: `EXT-X-STREAM-INF` per variant — `BANDWIDTH` (peak bit/s, required), `AVERAGE-BANDWIDTH`, `CODECS`, `RESOLUTION`, `FRAME-RATE`, group refs; `EXT-X-MEDIA` = alternate audio/subtitle renditions (`GROUP-ID`, `LANGUAGE`); `EXT-X-I-FRAME-STREAM-INF` = key frames only, for scrubbing thumbnails.
- Media** playlist: `EXT-X-TARGETDURATION`, `EXT-X-MEDIA-SEQUENCE`, `EXTINF` per segment. **VOD** ends with `EXT-X-ENDLIST`; **live** = sliding window the client re-fetches about every target duration; `EVENT` = append-only. `EXT-X-DISCONTINUITY` at an ad splice.
- Segments**: MPEG-TS or fMP4/CMAF + `init` (`EXT-X-MAP`). **Keys**: `EXT-X-KEY AES-128 / SAMPLE-AES`; `FairPlay` = `skd: // URL`.
- LL-HLS** (8216bis draft): **partial segments** (`EXT-X-PART`, sub-second), `PRELOAD-HINT`, **blocking reload** (`?_HLS_msn=...&_HLS_part=...` held until it exists; server says `CAN-BLOCK-RELOAD=YES`), `RENDITION-REPORT`, delta playlists (`EXT-X-SKIP`). Latency ≈ TV broadcast.

ABR — what you control

- The player estimates bandwidth from segment downloads + buffer health and picks the start variant itself; `startOnFirstEligibleVariant` (iOS 14) forces the first listed one.
- Caps on `AVPlayerItem`: `preferredPeakBitRate` (bit/s, 0 = none) · `...ForExpensiveNetworks` (iOS 15, cellular) · `preferredMaximumResolution`.
- `preferredForwardBufferDuration` (s; 0 = system decides): low = fast start, more stalls; high = memory + bytes wasted when the user scrolls away.

AVFoundation state — watch, don't poll

- `AVURLAsset` (media; try `await asset.load(.isPlayable, .duration)`) → `AVPlayerItem` (presentation: `status` · `.unknown/.readyToPlay/.failed, isPlaybackLikelyToKeepUp`, `loadedTimeRanges`, logs) → `AVPlayer` (rate, `timeControlStatus` · `.paused/.waitingToPlayAtSpecifiedRate/.playing`) → `AVPlayerLayer` · `AVPlayerViewController` · `SwiftUI` `VideoPlayer`.
- `reasonForWaitingToPlay`: `.toMinimizeStalls`, `.evaluatingBufferingRate`, `.noItemToPlay`, `.interstitialEvent`, `.waitingForCoordinatedPlayback`.
- `automaticallyWaitsToMinimizeStalling` (default **true**): `play()` may wait for buffer. `false` = start at once; an empty buffer then *pauses* (rate 0). Loops: `AVPlayerLooper` + `AVQueuePlayer`.

Example — observe, cap, measure

```
let item = AVPlayerItem(asset: AVURLAsset(url: hlsURL))
item.preferredForwardBufferDuration = 2 // feed: start fast
item.preferredPeakBitRateForExpensiveNetworks = 1_500_000
let player = AVPlayer(playerItem: item)
obs = [item.observe(.status) { it, _ in
    if it.status == .failed { report(it.error) },
    player.observe(.timeControlStatus) { p, _ in
        spin(p.timeControlStatus == .waitingToPlayAtSpecifiedRate) },
    layer.observe(.isReadyForDisplay) { l, _ in // hide poster
        poster.isHidden = l.isReadyForDisplay }
NotificationCenter.default.addObserver(forName: .AVPlayerItemNewAccessLogEntry,
    object: item, queue: .main) { _ in
        guard let e = item.accessLog()?.events.last else { return }
        qoe.send(e.startupTime, e.numberOfStalls, e.indicatedBitrate) }
    player.play()
}
```

Android: `Media3 ExoPlayer` (HlsMediaSource, DASH too), `TrackSelectionParameters` caps, `LoadControl` buffers, `MediaSessionService` for background, Widevine DRM.

Startup time and feeds

- Start fast**: create the asset/item *before* the tap (prefetch the next cells), load keys async, short buffer target, a low first variant, show a **poster** until `isReadyForDisplay`. Measure time-to-first-frame, not `play()` returning.
- Feeds**: a **pool** of 2–3 `AVPlayers`, not one per cell; swap with `replaceCurrentItem(with:)`; only the most-visible cell plays (muted); set off-screen items to `nil` — each item holds buffers and a decoder. Cap resolution to the cell's pixels.

Background, PiP, AirPlay

- Background audio**: `AVAudioSession.playback + UIBackgroundModes: audio`. For a video, `playerLayer.player = nil` on background (QA1668), re-attach on foreground.
- PiP**: `AVPictureInPictureController`, same mode; `canStartPictureInPictureAutomaticallyFromInline` (14.2). `AVPlayerViewController`: free.
- AirPlay**: `AVRoutePickerView`; with `allowsExternalPlayback` the Apple TV fetches the stream itself — its URLs and auth must work from there.

DRM, offline, metrics

- FairPlay**: `AVContentKeySession(keySystem: .fairPlayStreaming)` — key request → `makeStreamingContentKeyRequestData = SPC` → your key server → `CKC` → `processContentKeyResponse`. Offline needs a *persistable* key.
- Offline**: `AVAssetDownloadURLSession` + `makeAssetDownloadTask(downloadConfiguration:)`; the system picks the location — store the URL it returns.
- QoE**: `accessLog()` events (`indicatedBitrate`, `observedBitrate`, `numberOfStalls`, `startupTime`, `numberOfDroppedVideoFrames`) + `errorLog()`; not KVO — use the notifications. KPIs: startup time, rebuffer ratio, bitrate, fatal errors.

Traps

- ! Waiting on `AVPlayer.status` — it's the *item's* status that means ready.
- ! Spinner from `rate == 0` — rate is the *requested* speed; use `timeControlStatus`.
- ! New `AVPlayer` per cell: memory, decoders, hitches. One item in two players: exception.
- ! KVO token not stored → no callbacks; time observer never removed → leak.
- ! `isPlaybackLikelyToKeepUp` can be false with a *full* buffer — it's a prediction.

Remember

Playlist of playlists of segments · the player picks, you *cap* · watch the *item* for ready, the *player* for waiting · pool players in feeds.